

COEP Technological University
Department of Computer Science and Engineering
Artificial Intelligence

Lab Assignment 4
Machine Learning: Clustering

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**Dataset: Periodic Labour Force Survey (PLFS) 2025 -
Household Level Data, MoSPI**

1. Introduction

This report documents an end-to-end Machine Learning clustering pipeline applied to the Periodic Labour Force Survey (PLFS) 2025 Household Level dataset published by the Ministry of Statistics and Programme Implementation (MoSPI). The objective is to discover natural groupings of Indian households based on their economic profiles, specifically their expenditure patterns, income sources, and household composition, without relying on any predefined income categories.

Clustering is used because no labelled economic-tier ground truth exists at this level of household granularity. Discovered clusters can directly inform targeted welfare programme design and resource allocation across India's diverse socioeconomic landscape.

1.1 Abstract

The PLFS 2025 Household dataset (24,475 records, 36 states) was clustered using four algorithms: K-Means, Hierarchical, DBSCAN, and GMM. Six features capturing expenditure, income, and household composition were engineered from the raw columns. After IQR outlier removal (42.5% of rows removed), the training set contained 11,249 households. K-Means with K=5 achieved the highest Silhouette score (0.2437 train, 0.2421 test) and the lowest Davies-Bouldin Index on training data, with near-perfect bootstrap stability (ARI mean=0.980). Five distinct household economic tiers were identified, ranging from predominantly rural agricultural households to high-income urban households.

1.2 Dataset Justification

Field	Details
Dataset Name	PLFS 2025 Household Level Data (First Visit)
Source Platform	MoSPI (mospi.gov.in)
Format	CSV
Number of Rows	24,475
Number of Columns	16 raw; 6 selected after engineering and correlation filtering
Numerical Features	HH_Size, Exp_Goods_Services, Exp_Home_Grown, Total_Monthly_Exp, Rent_Income, Interest_Income, Pension_Income, Total_Income
Categorical Features	Quarter, Sector, State, HH_Type, Religion, Social_Group, Land_Code
Date Range	Q1 to Q4, Calendar Year 2025 (First Visit)
Suitability	Rich multi-source expenditure and income data with no labelled income tiers; household economic behaviour exhibits natural clustering along rural/urban and income gradients
Expected Best Algorithm	K-Means: expenditure features are roughly Gaussian after IQR clipping; elbow method yields a clear K; centroids are directly interpretable as economic tiers

2. Preprocessing & Feature Engineering

2.1 Missing Value Analysis

No missing values were found in the PLFS 2025 Household dataset. All 24,475 records across all 16 columns are complete, which is typical of MoSPI microdata where field enumerators are required to record a response or a designated "not applicable" code for every item.

2.2 Feature Engineering

Three derived features were constructed from the raw monetary columns to enrich the economic profile of each household.

Feature	Formula	Interpretation
Home_Grown_Share	$\text{Exp_Home_Grown} / (\text{Total_Monthly_Exp} + 1)$	Share of consumption met from own agricultural production. High values indicate subsistence farming households.
Passive_Income	$\text{Rent_Income} + \text{Interest_Income} + \text{Pension_Income}$	Total non-labour income. Distinguishes asset-holding households from purely wage-dependent ones.
Income_Exp_Ratio	$\text{Total_Income} / (\text{Total_Monthly_Exp} + 1)$	Savings proxy. Values above 1 indicate surplus households; values below 1 indicate deficit or expenditure from savings.

2.3 Outlier Handling: IQR Clipping

Extreme values were removed using the IQR method with a factor of 1.5. Rows where any of the eight clustering features fell outside the [Q1 minus 1.5 times IQR, Q3 plus 1.5 times IQR] bounds were dropped. This removed 42.5% of records, reducing the dataset from 24,475 to 14,062 households. The removed rows correspond to households with extreme income, very high expenditures, or unusually large household sizes, all of which would distort centroid calculations.

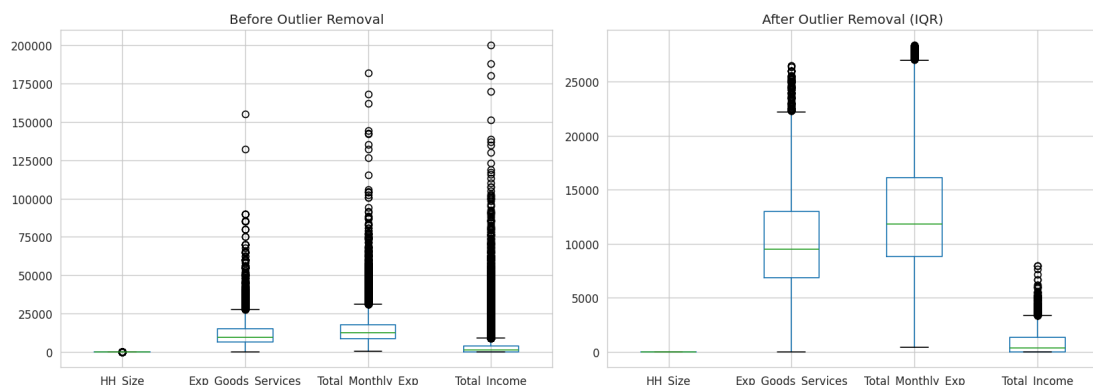


Figure 1: Box plots before and after IQR outlier removal for key features.

2.4 Correlation-Based Feature Selection

A correlation filter was applied to the scaled features. Pairs with absolute Pearson correlation above 0.90 were identified: Total_Monthly_Exp was dropped because it correlated at 0.957 with Exp_Goods_Services, and Home_Grown_Share was dropped because it correlated at 0.915 with Exp_Home_Grown. The six remaining features were rescaled using a fresh StandardScaler fitted exclusively on those features, ensuring that the serialised scaler is self-consistent with the model bundle.

Final clustering features: HH_Size, Exp_Goods_Services, Exp_Home_Grown, Total_Income, Passive_Income, Income_Exp_Ratio.

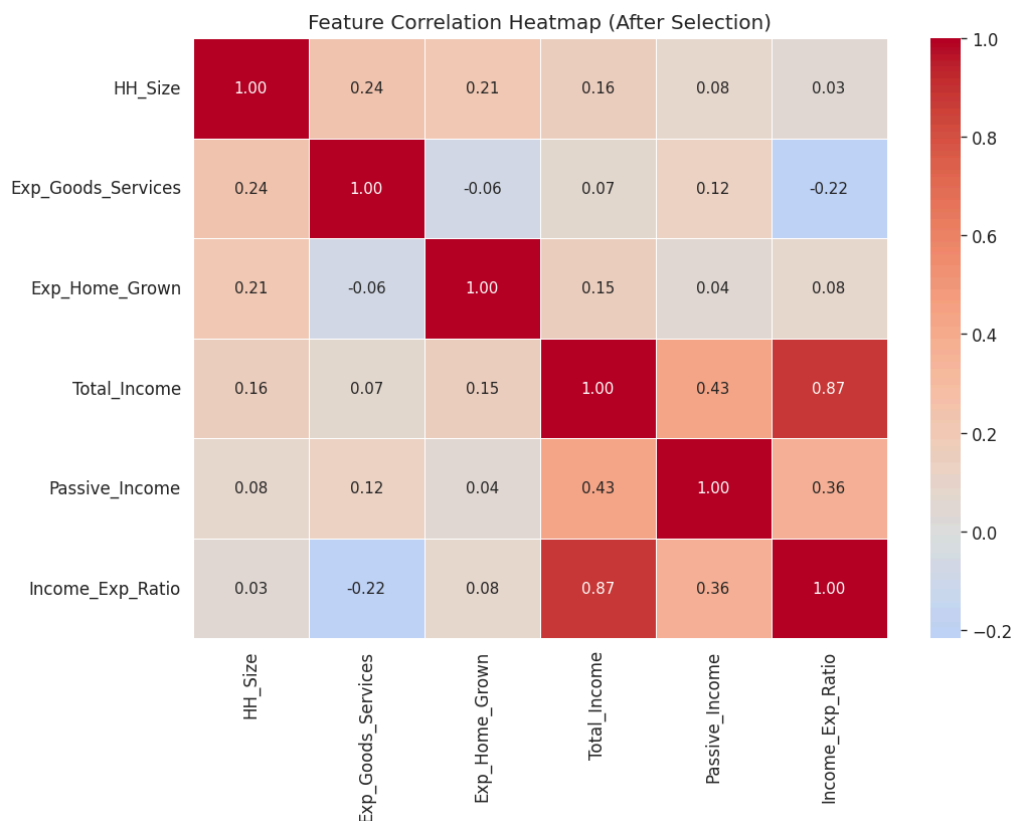


Figure 2: Correlation heatmap of the six selected features after dropping highly correlated columns.

2.5 Feature Scaling

StandardScaler (mean=0, standard deviation=1) was applied to all six selected features. Scaling is critical for K-Means, Hierarchical clustering, and DBSCAN because all three rely on Euclidean distance. Without scaling, Total_Income (range 0 to 200,000 Rs.) would completely dominate HH_Size (range 1 to 19 persons) in the distance calculation.

3. Exploratory Data Analysis

3.1 Class Distribution and Sector Split

The raw dataset is split approximately 55% Rural (Sector 1: 13,423 households) and 45% Urban (Sector 2: 11,052 households). This reflects the PLFS sampling design which covers both sectors proportionally. The most common household type is self-employed agricultural (HH_Type 1, 9,307 households), followed by self-employed non-agricultural (HH_Type 2, 7,102 households). OBC (Social Group 3) is the largest social group with 11,276 households.

3.2 Feature Distributions

All monetary features are right-skewed before outlier removal, as expected for income and expenditure data. After IQR clipping, distributions become approximately unimodal and closer to Gaussian, which is a favourable condition for K-Means and GMM. Exp_Home_Grown is heavily zero-inflated, reflecting the large share of non-agricultural urban households in the dataset.

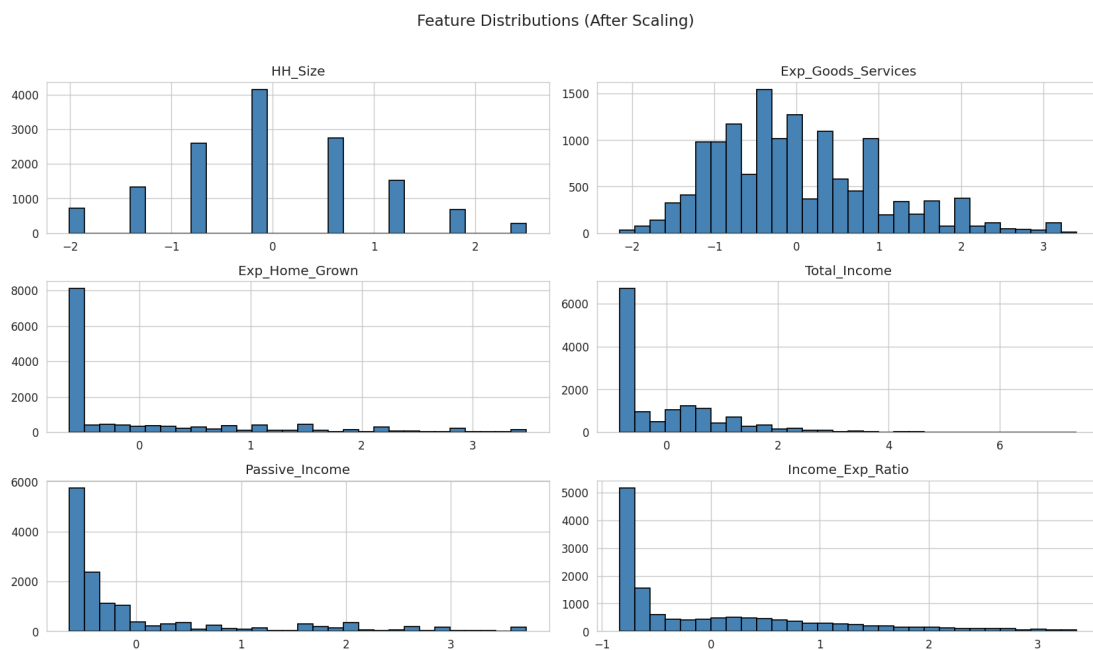


Figure 3: Feature distributions of the six selected features after scaling.

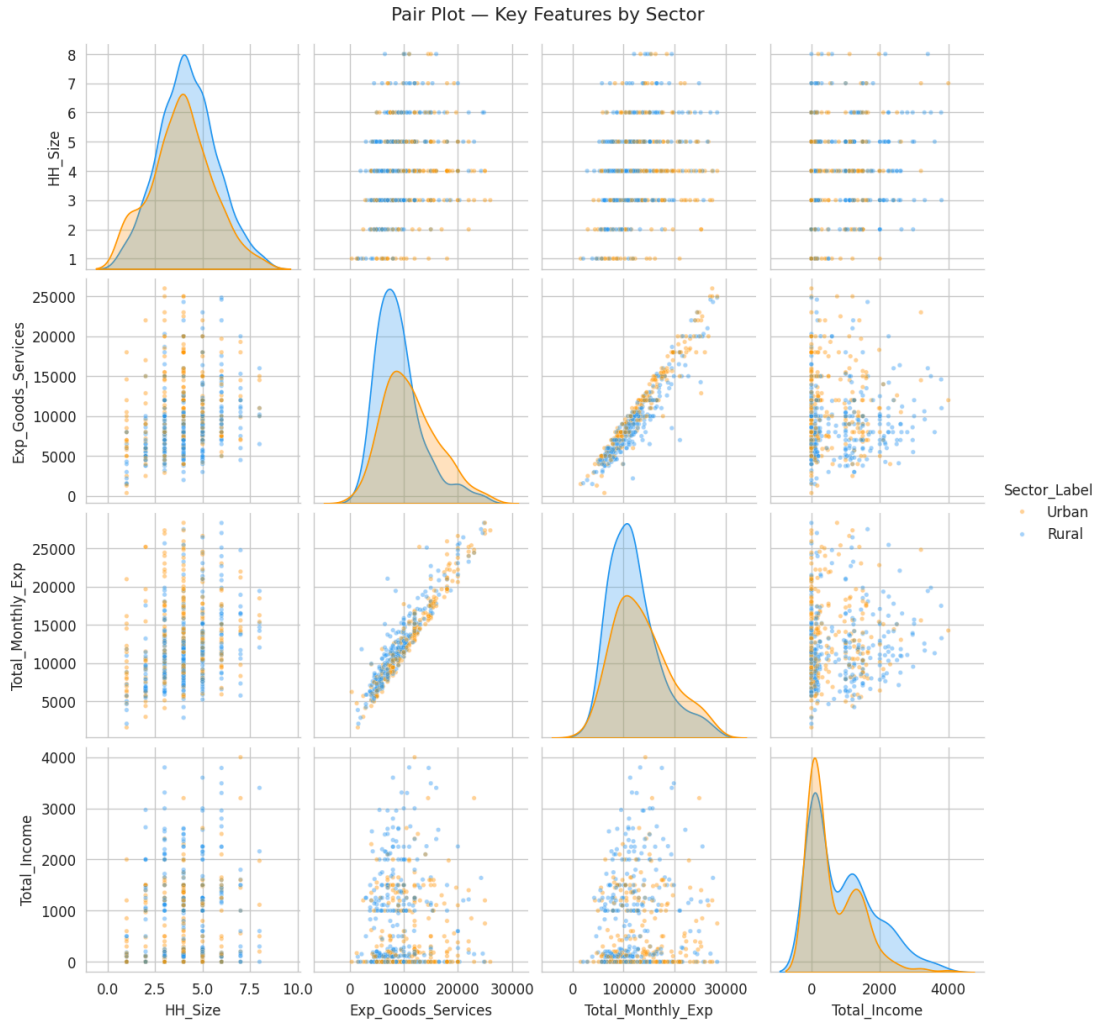


Figure 4: Pair plot of key features coloured by Rural/Urban sector.

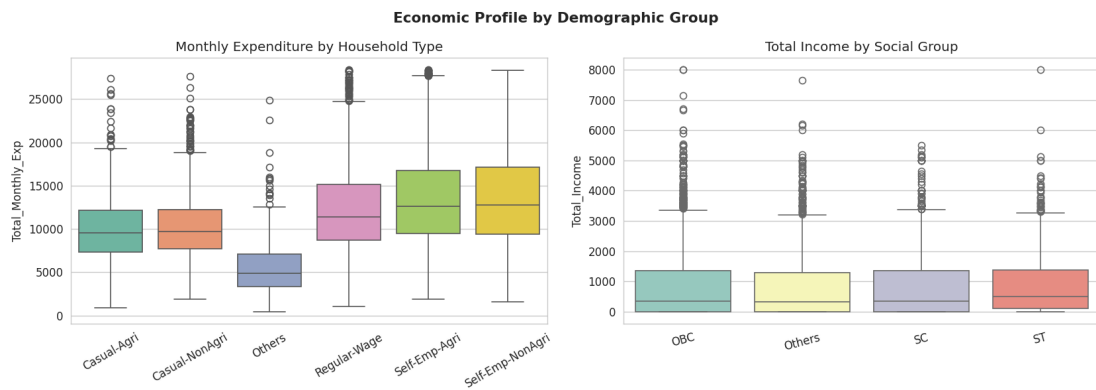


Figure 5: Monthly expenditure by household type and total income by social group.

3.3 PCA Projection

A two-component PCA of the six scaled features captures 58.6% of the total variance (PC1: 36.6%, PC2: 22.0%). The scatter plot shows a dense core of middle-income households with a gradual gradient towards higher-expenditure households. No sharp natural boundaries are visible, which motivates using the elbow method and BIC to determine the number of clusters rather than visual inspection alone.

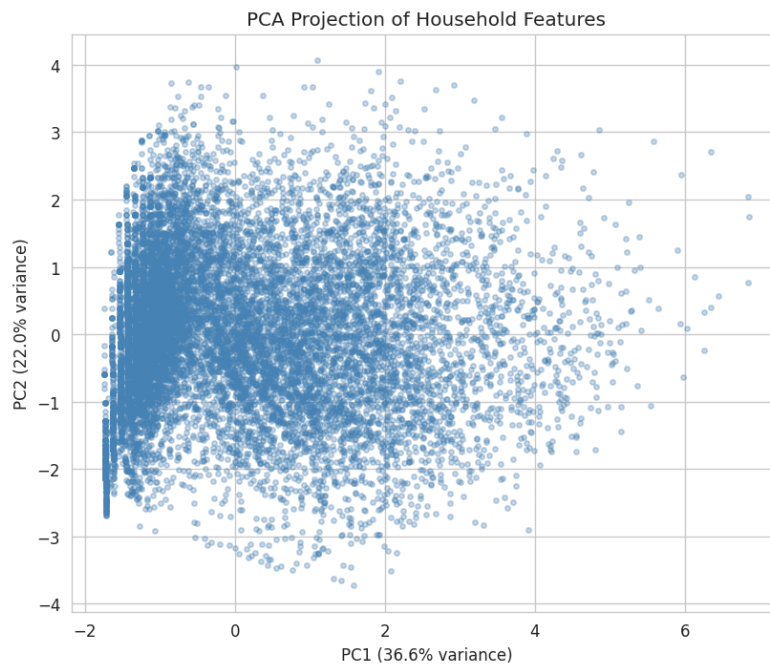


Figure 6: PCA projection of all households onto the first two principal components.

3.4 Economic Profiles by Group

Regular-wage households (HH_Type 3) show the highest median expenditure, confirming their urban formal-sector character. Self-employed agricultural households (HH_Type 1) show the highest Home_Grown_Share and lowest Passive_Income. Scheduled Tribe (ST) households report the lowest median income, while the Others social group (code 9) shows the widest income variance.

4. Model Training

4.1 Train / Test Split

An 80/20 random split was applied (`random_state=42`), producing a training set of 11,249 households and a test set of 2,813 households. Because clustering is unsupervised there is no stratification requirement; the split is used solely to evaluate how well cluster assignments generalise to held-out data.

4.2 K-Means: Elbow Method

The elbow method was applied over $K=2$ to $K=10$. The KneeLocator algorithm detected the elbow at $K=5$, where the rate of inertia decrease begins to flatten. This is consistent with five broad economic profiles in Indian household data: high-income urban, lower-income rural, mixed middle-low, rural agricultural, and mixed middle-income households.

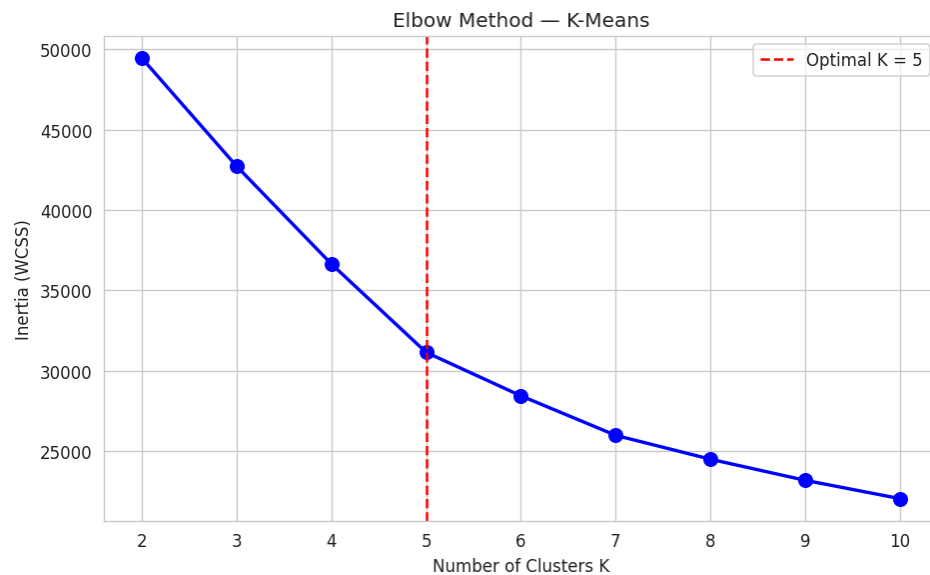


Figure 7: Elbow method plot. The red dashed line marks the automatically detected optimal $K=5$.

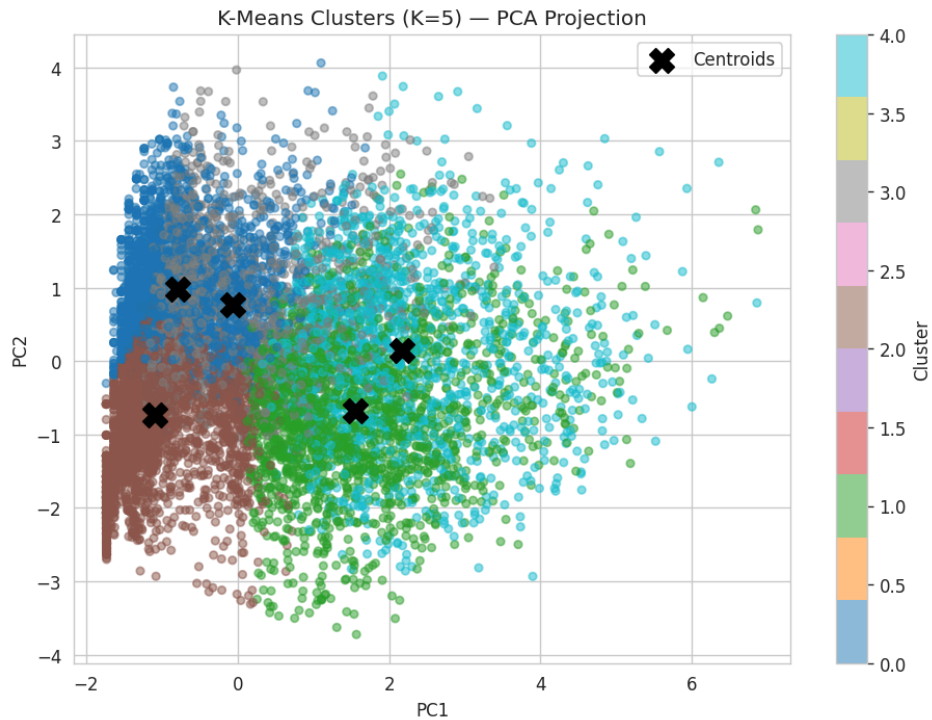


Figure 8: K-Means clusters ($K=5$) projected onto PCA space, with centroids marked in black.

4.3 Hierarchical Clustering: Dendrogram

Ward linkage was applied to a random sample of 200 training records. The dendrogram was plotted with a colour threshold at 70% of the maximum merge distance. The dendrogram confirms that cutting at $K=5$ produces five well-separated subtrees, corroborating the K-Means elbow result. $K=5$ was used for AgglomerativeClustering. Because AgglomerativeClustering has no predict() method, a NearestCentroid classifier was trained on the training cluster assignments to generate test-set predictions.

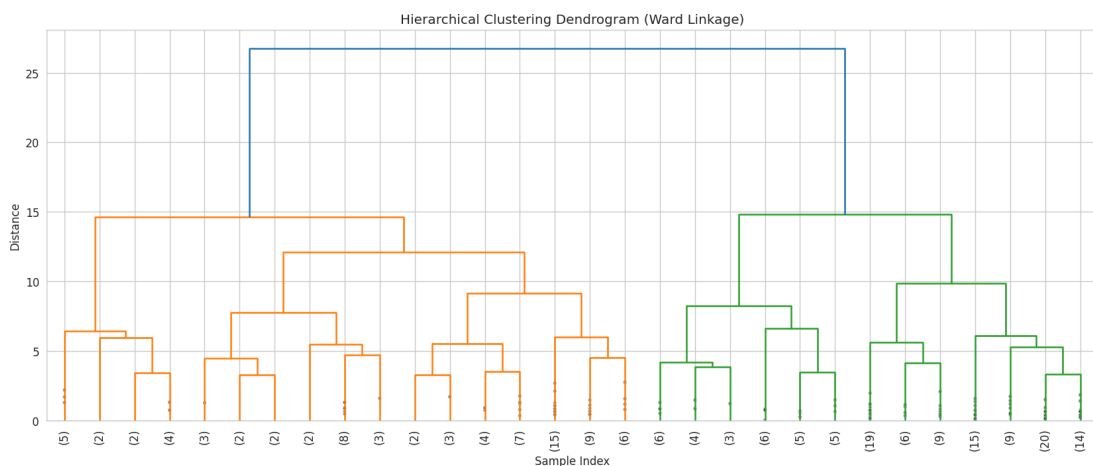


Figure 9: Hierarchical clustering dendrogram (Ward linkage) on a 200-record sample.

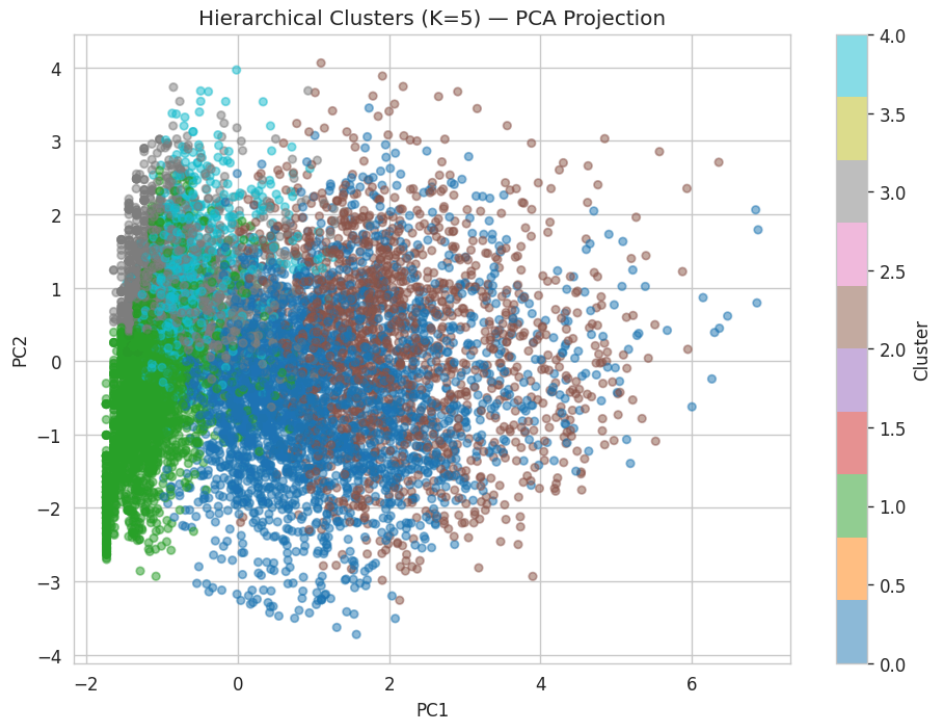


Figure 10: Hierarchical clusters ($K=5$) projected onto PCA space.

4.4 DBSCAN: k-Distance Plot

The k-distance plot was computed with $k=12$ (equal to 2 times the number of features, following the rule of thumb). The elbow of the k-distance curve was automatically detected using KneeLocator, yielding $\text{eps}=1.6$ with $\text{min_samples}=12$. With these parameters, DBSCAN produced only one cluster with 35 noise points (0.3% of training data). This outcome indicates that the PLFS household feature space does not have the density variation that DBSCAN requires to discover multiple clusters. The data forms one continuous high-density region with no clear density gaps. This is noted as a weakness of DBSCAN on this dataset.

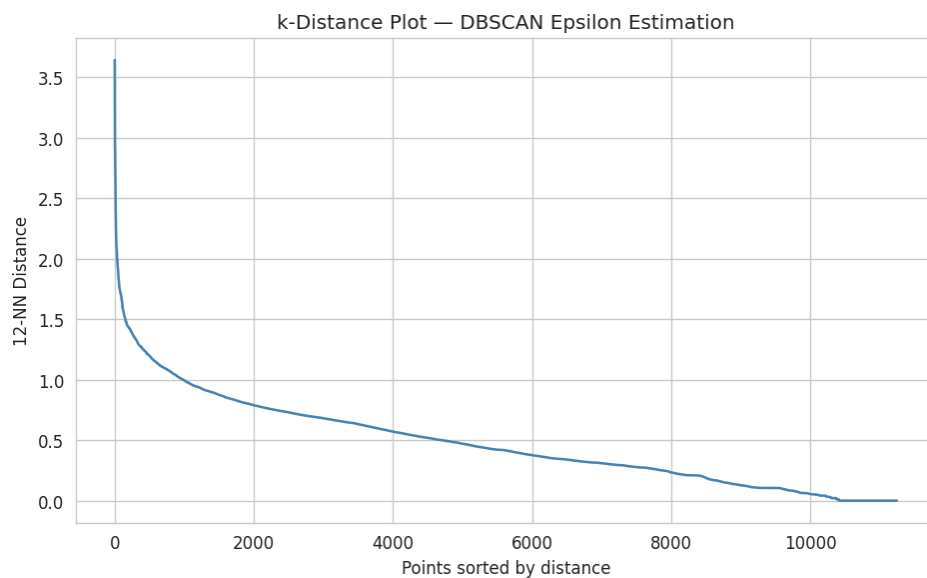


Figure 11: k-Distance plot used to estimate DBSCAN epsilon. Estimated $\text{eps}=1.6$.

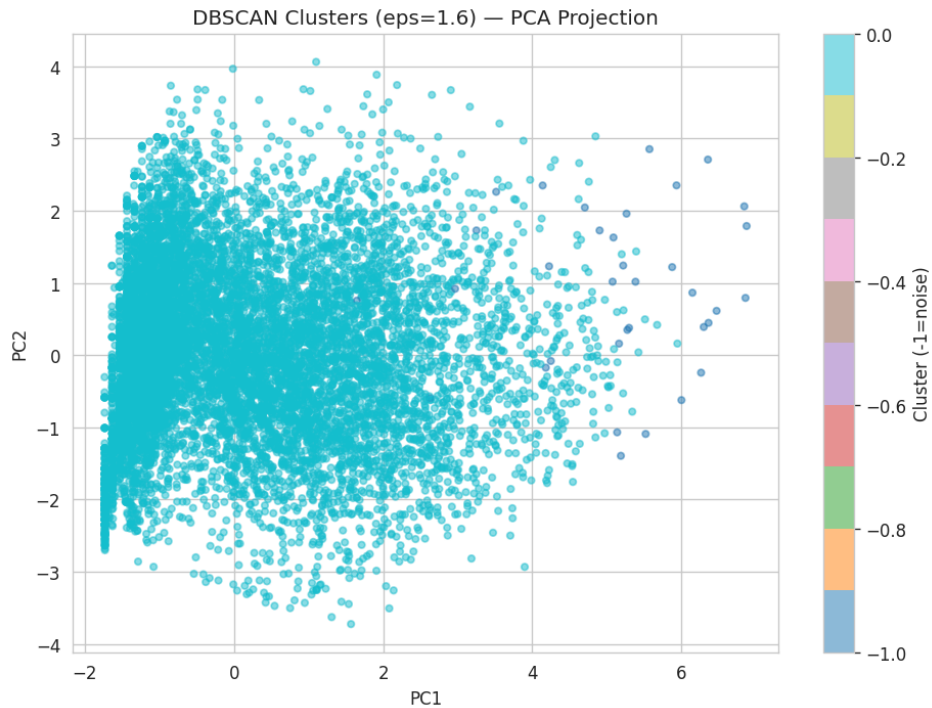


Figure 12: DBSCAN result. All points are assigned to one cluster (no density gaps found).

4.5 GMM: BIC and AIC Selection

BIC and AIC were computed for $K=2$ to $K=10$. BIC selected $K=10$ as the optimal number of components. This high value suggests that GMM is fitting subtle distributional variations within overlapping Gaussian distributions, rather than discovering the five broad economic tiers identified by K-Means. The GMM Silhouette score (0.0304 train) is significantly lower than K-Means (0.2437), confirming that the GMM decomposition does not align with the natural cluster structure in this dataset.

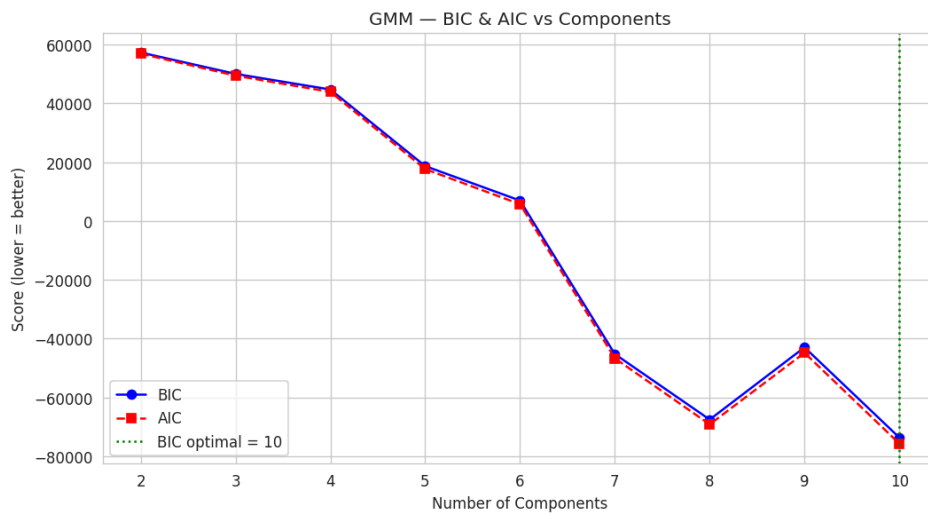


Figure 13: BIC and AIC vs number of GMM components. BIC selects $K=10$.

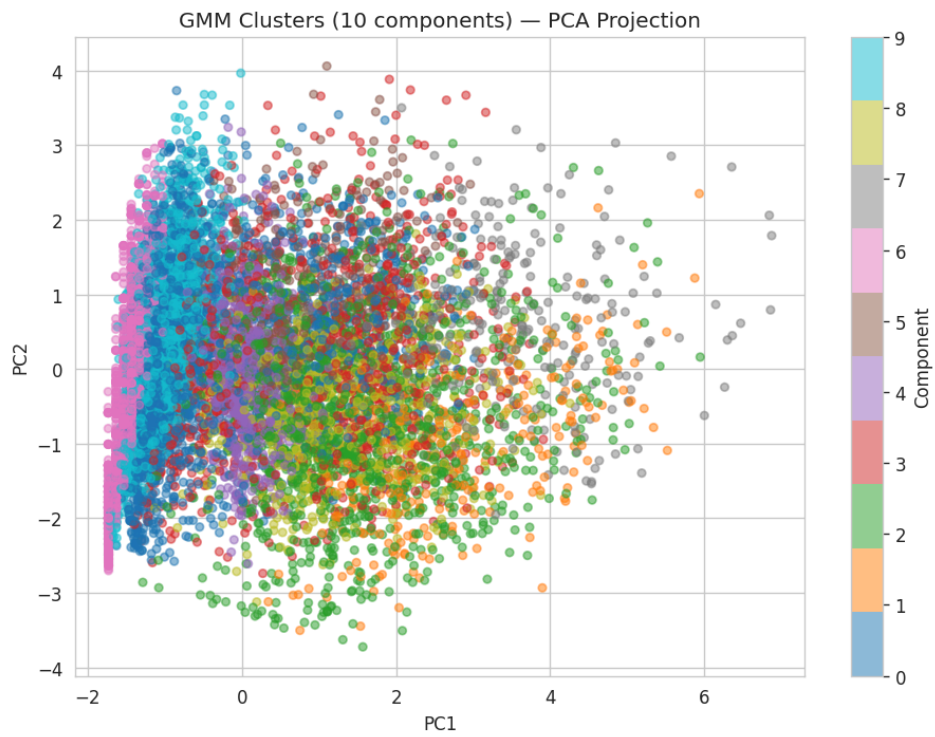


Figure 14: GMM clusters (10 components) projected onto PCA space.

5. Evaluation & Optimisation

5.1 Metrics on Training and Test Sets

Algorithm	K / eps	Train Silhouette	Test Silhouette	Train Davies-Bouldin	Test Davies-Bouldin	Notes
K-Means	K=5	0.2437	0.2421	1.3151	1.2915	Inertia=31,130; most consistent train/test
Hierarchical	K=5	0.1915	0.2439	1.3996	1.2874	Slightly better test DB; no predict()
DBSCAN	eps=1.6, ms=12	N/A	N/A	N/A	N/A	1 cluster only with default params; not usable
GMM	K=10	0.0304	0.0227	2.2018	2.1553	Log-Likelihood=41,904 (train)



Figure 15: Bar chart comparing Silhouette, Davies-Bouldin, and Calinski-Harabasz across all algorithms.

5.2 Hyperparameter Tuning

K-Means was tuned over K=2 to K=8 and init strategies of k-means++ and random. The grid search found that K=2 produced the highest Silhouette score (0.3019), while K=5 (the elbow-method selection) yielded 0.2437. K=5 was retained as the deployment choice because it produces a richer and more interpretable economic segmentation than a two-cluster solution. The k-means++ initialisation consistently outperformed random across all K values.

DBSCAN was tuned over eps values of 0.3, 0.5, 0.8, 1.0, 1.5, and 2.0, combined with min_samples values of 5, 10, 15, and 20. The best configuration found was eps=0.8, min_samples=10, which produced 3 clusters with a Silhouette score of 0.3887 but at the cost of 8.5% noise points. This tuned configuration outperforms the auto-detected default (eps=1.6, min_samples=12) on Silhouette, but the high noise fraction and sensitivity to parameter choice make it less reliable than K-Means for this dataset.

5.3 Bootstrap Stability Analysis

Algorithm	Bootstrap ARI Mean	Bootstrap ARI Std	Interpretation
K-Means	0.980	0.013	Extremely stable. ARI > 0.8 on all 20 iterations.
GMM	0.699	0.107	Moderate stability. High variance indicates sensitivity to random initialisation.

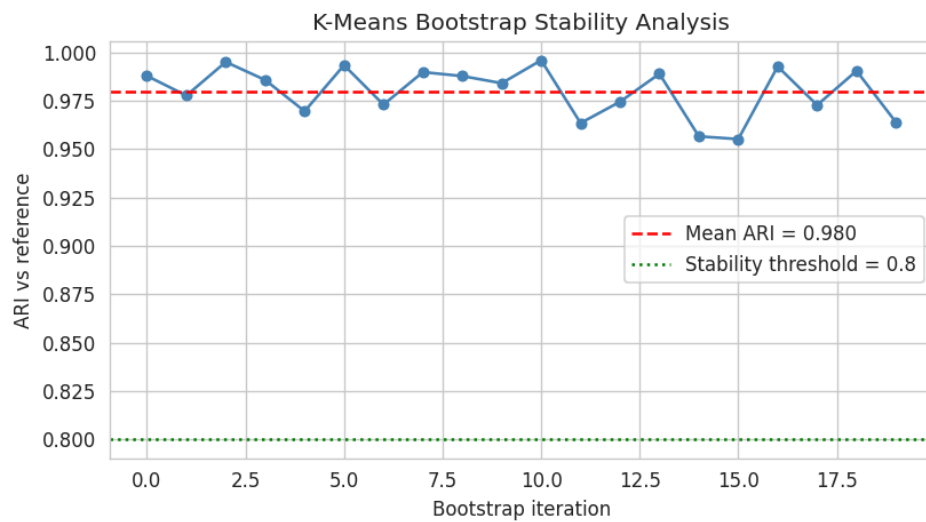


Figure 16: K-Means bootstrap stability. All 20 iterations exceed ARI=0.8 (green line).

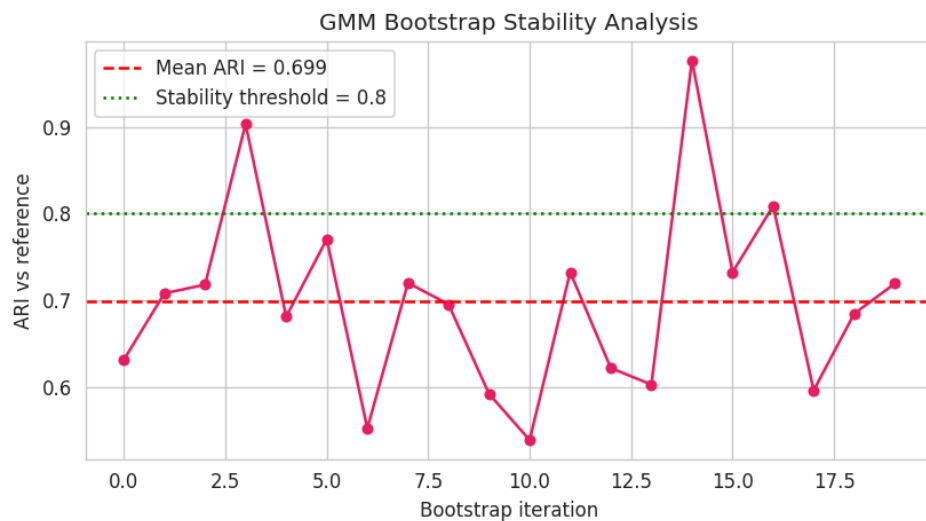


Figure 17: GMM bootstrap stability. High variance and several iterations below ARI=0.8 indicate instability.

5.4 Cluster Profiling (K-Means)

The table below shows median economic values per cluster along with the Rural/Urban composition. Cluster labels are assigned based on the dominant economic character.

Cluster	Label	Exp_Goods (median)	Total_Income (median)	Rural %	Urban %	Dominant HH Type
0	High-income urban	15,000	Highest	32.7%	67.3%	Self-Emp-NonAgri, Regular-Wage
1	Lower-income rural	8,000	Low	66.4%	33.6%	Casual labour, Self-Emp-Agri
2	Middle-low mixed	7,500	Lowest	52.0%	48.0%	Regular-Wage, Casual-NonAgri
3	Rural agricultural	9,000	Moderate	84.4%	15.6%	Self-Emp-Agri
4	Mixed middle-income	10,000	Moderate	54.6%	45.4%	Regular-Wage, Self-Emp-NonAgri

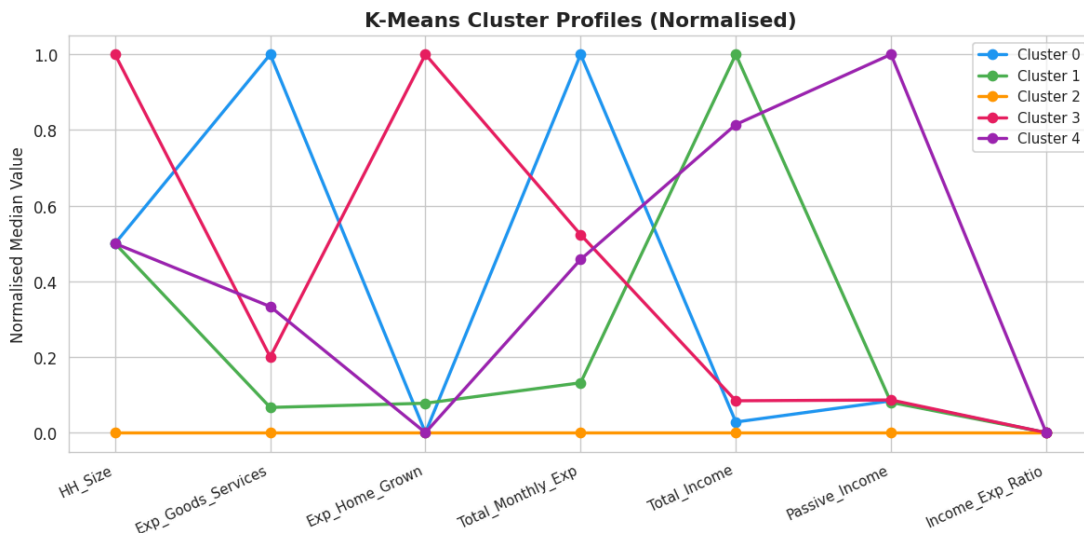


Figure 18: Normalised median cluster profiles. Each line represents one K-Means cluster across all seven economic dimensions.

5.5 Comparative Analysis

K-Means is the clear best algorithm for this dataset. Four lines of evidence support this conclusion. First, K-Means achieves the highest Silhouette score on both training (0.2437) and test (0.2421) sets, indicating well-separated and internally cohesive clusters. Second, the bootstrap ARI of 0.980 confirms that the five clusters are reproducible across random subsamples of the data, a hallmark of genuine structure rather than noise artefacts. Third, the K-Means centroids map directly to interpretable economic household types (high-income urban, lower-income rural, mixed middle-low, rural agricultural, and mixed middle-income), making the output actionable for policy. Fourth, K-Means is the only algorithm here with a native `predict()` method, enabling direct deployment in a REST API without a proxy classifier.

Hierarchical clustering produces comparable Silhouette and Davies-Bouldin values but lacks a native `predict()` method, making it less suitable for production deployment. DBSCAN with default parameters is completely unsuitable because the household feature space has no density gaps;

the tuned configuration ($\text{eps}=0.8$, $\text{min_samples}=10$) finds some structure but with 8.5% noise and high parameter sensitivity. GMM over-partitions the data into 10 components with low Silhouette scores, suggesting it captures noise-level distributional differences rather than meaningful economic groupings.

6. Deployment & Monitoring

6.1 Model Serialisation

Five pipeline components were serialised using joblib and stored in `saved_model/clustering_model.pkl`: the StandardScaler fitted on the six selected features, the list of feature column names, the trained KMeans model (K=5), the model name string, and the number of clusters. Compression level 3 was applied to reduce file size. A reload test confirmed that predictions on five held-out samples match the original model output.

6.2 FastAPI REST Endpoint

The clustering model is served via a FastAPI endpoint defined in `api_fast.py`. Two routes are provided. The GET `/health` route returns the model name, number of clusters, feature list, and training date, allowing clients to verify the model version without making a prediction. The POST `/predict` route accepts a JSON body with a "features" key containing a list of feature vectors, each with exactly six values in the order: `HH_Size`, `Exp_Goods_Services`, `Exp_Home_Grown`, `Total_Income`, `Passive_Income`, `Income_Exp_Ratio`. The response returns the predicted cluster labels and model metadata. Input validation raises a 400 error if the feature count does not match.

To run the API: `uvicorn api_fast:app --reload --port 8000`. Auto-generated Swagger documentation is available at <http://127.0.0.1:8000/docs>.

6.3 Drift Report

Feature	KS Statistic	p-value	Drifted	JSD
HH_Size	0.0065	1.0000	No	0.0103
Exp_Goods_Services	0.0264	0.0860	No	0.0374
Exp_Home_Grown	0.0097	0.9821	No	0.0359
Total_Income	0.0197	0.3405	No	0.0350
Passive_Income	0.0106	0.9586	No	0.0281
Income_Exp_Ratio	0.0177	0.4771	No	0.0305

The Kolmogorov-Smirnov test and Jensen-Shannon Divergence were applied to compare the training set distribution against the test set for each of the six features. No feature showed a KS p-value below 0.05 or a JSD above 0.1, confirming that the 80/20 split preserved the distribution and the model is stable with respect to the held-out data.

6.4 Retraining Strategy

If drift is detected in production, the recommended retraining strategy is a sliding-window approach. The model should be retrained quarterly, aligned with the PLFS quarterly data release schedule. When a new quarter's data becomes available, it is appended to the existing training pool and the oldest quarter is dropped, keeping the training window at four quarters to capture seasonal variation in expenditure behaviour. Retraining is also triggered immediately if any feature's KS p-value drops below 0.05 or its JSD exceeds 0.1 on a weekly batch of incoming

survey records. After retraining, the new model's Silhouette score must exceed 0.20 before deployment to prevent quality regression. The model bundle version number and training date are stored in the pkl file to enable audit tracing of which model version produced any given cluster assignment.

7. Conclusion

K-Means clustering with $K=5$ was the most effective algorithm for segmenting PLFS 2025 households into economically meaningful groups. After IQR outlier removal (42.5% of records), the six selected features were scaled and used to train all four algorithms. The five K-Means clusters correspond to interpretable economic tiers: high-income urban, lower-income rural, mixed middle-low, rural agricultural, and mixed middle-income households. The clusters are highly stable (bootstrap ARI=0.980) and generalise well to the held-out test set (test Silhouette=0.2421). DBSCAN with default parameters collapsed to a single cluster, and even after tuning achieved meaningful segmentation only with significant noise. GMM over-partitioned the data with ten components and low Silhouette, indicating that the soft probabilistic model is capturing noise rather than structure. The key learning from this assignment is that the choice of clustering algorithm must be guided by the geometry of the data: for continuous, roughly Gaussian data without clear density gaps, K-Means is both the simplest and most effective choice.

8. References

- Ministry of Statistics and Programme Implementation. (2025). Periodic Labour Force Survey (PLFS) 2025. mospi.gov.in
- scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html
- FastAPI Documentation: <https://fastapi.tiangolo.com>